

FLG: A Forest Landscape Generator

by

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Introduction

With increased attention to non-timber forest ecosystem values, there is a growing need for spatially referenced models to optimize resource allocation as part of an Integrated Forest Resource Management (IFRM) strategy. For example, patch distribution and age class structure are often used to indirectly represent biodiversity, habitat, and visual quality of landscapes. The associated forestry management decision problems are to choose the best forest management intervention strategies so as to create desired spatial structures in forests. These forestry operations research problems have provided interesting and difficult challenges to operational researchers.

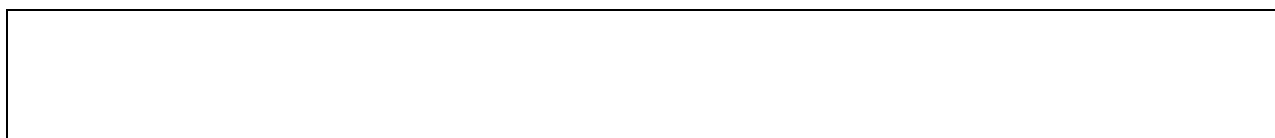
Optimization models for these problems typically have non-linear objective functions and constraints, involve multiple decision criteria and are often solved using heuristic methods. Results are often shown for a single case study forest with static spatial conditions. This has led to challenges in comparing methods proposed by different researchers. As well, to demonstrate robustness and performance of any given algorithm, it is necessary to quantify the sensitivity of model output and performance to variations in the spatial distribution of forest. To complete such a sensitivity analysis, it is necessary to have available a sample of 'forests' which vary in a controlled random manner. Algorithm developers can benefit from testing their methods on samples of forests with similar spatial structures or on forest configurations that cover a range of initial spatial conditions.

Although modeling approaches and user interfaces vary widely, almost all make use of a Geographic Information System (GIS) database to define the size, shape and relative positions of the patches that make up the forest. It is relatively easy for a user to modify feature attributes in a GIS database, and hence to create samples of forests with varied initial forest conditions. That is, it is easy to do so without changing the size, shape and relative positions of the patches that make up the GIS database. It is not so easy to create forests with different spatial arrangements of patches, while controlling for the distribution of patch sizes, vegetation types, and ages. Manual methods, where the user "shuffles" the landscape around interactively, are very likely to introduce bias into the spatial arrangement of the patches.

This article describes the design and implementation of the *Forest Landscape Generator* (FLG) – a GIS based software tool that generates sample forests for model testing purposes.

Initial Design of FLG

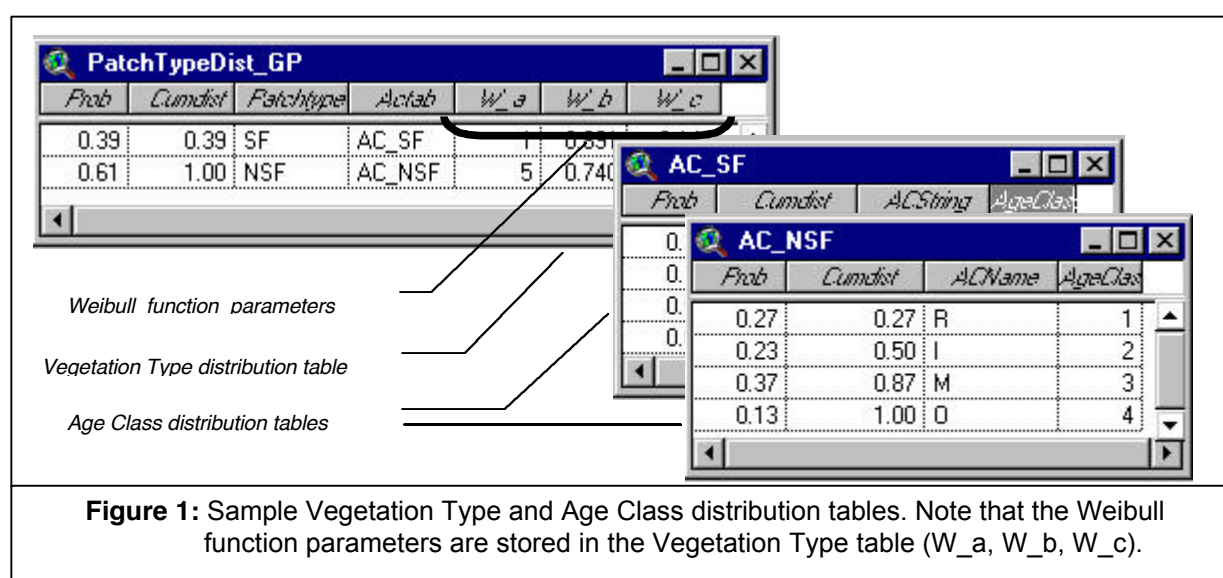
The initial design for the FLG was based on a senior thesis project (Gary Whittaker, Forest Engineering, UNB, 1994). The Whittaker model generates a rectangular matrix of equal-area hexagonal cells representing a forest of a user-defined length and width. Patches are generated in a



pseudo-random fashion by selecting an ageclass from a discrete age distribution, selecting an unallocated starting cell (mother), generating a patch size from a Weibull distribution (number of daughters), and then allocating the required number of daughter cells to create the patch. Daughter cells are spatially distributed around the mother cell using a list of unallocated cells sorted by distance from the mother. The method thus attempts to create a contiguous patch for each randomly generated patch size. The FLG system extends on this basic method by improving on the patch growth algorithm. It allows for multiple distributions of patch size by vegetation type, and is implemented in a more user-friendly GIS environment.

FLG Design

The FLG was written using the Avenue™ scripting language provided with ArcView™ GIS, and uses



a raster-based landscape generation algorithm. The user may create, store, and edit all distributions as table documents within the same ArcView™ project used to run the FLG. Run-time user input for the FLG consists of cell size, matrix length and width, and XY coordinate offset for the lower left corner of the model. The vegetation type distribution is stored in a database table. Each vegetation type has an ageclass distribution and patch size distribution. Weibull parameters for the patch size distribution function are stored in the vegetation type database table.

The system produces an ArcView™ GRID raster file, and optionally a dissolved vector polygon shapefile. A script to generate adjacency lists for patches is included in the system.

Landscape Features and Distributions

The FLG allows the user to specify patch size and ageclass distributions for any number of vegetation types. The three-parameter Weibull distribution is used to model patch sizes, and tabular cumulative probability distributions for ageclasses and vegetation types. The Weibull distribution has been widely used to model the distributions of stand descriptors such as basal area and stem diameter. Studies in both Ontario and New Brunswick have shown that forest patch sizes demonstrate an exponential, or “inverse-J” distribution, which is easily modeled by the Weibull distribution. This function is very flexible – manipulation of the three parameters can produce a wide

variety of unimodal continuous patch size distributions. Moreover, the Weibull distribution function is easily inverted for Monte Carlo sampling.

To emulate an existing forest, the user would first stratify the forest by vegetation type, and observe the distribution of the number of patches in each stratum. Then, Weibull parameters for each vegetation type are found by fitting data from the existing forest patches. For each vegetation type, the user chooses ageclass definitions and calculates their distribution. Once stored in appropriately formatted database tables (see Figure 1), these distributions can be used to generate any number of landscapes with the FLG.

Spatial Allocation Of Patches

Although the daughter count samples generated by the Whittaker algorithm fit the user-defined Weibull distribution, the resulting landscape was more fragmented than was specified by the input distribution.

The FLG uses an iterative spatial allocation algorithm that limits patch fragmentation. For each patch, a ‘mother’ cell is randomly selected. Then, successive layers of unallocated cells around the mother cell are selected until there are enough cells to contain all daughters. The cells are then assigned appropriate vegetation and age attributes. Since adjacent cells are selected in layers, there will almost always be cells on the outer layer which are not assigned to the patch. To reduce spatial bias in the outer shape, cells are randomly selected from the outer layer.

This approach allows patches to “squeeze” through unallocated corridors while being boxed in by previously allocated cells. Patch growth is halted if the model runs out of adjacent unallocated cells, even if the patch is too small to contain the prescribed number of daughter cells. To reduce the effect of truncating the patch in this manner, we generate a random sample of patch attribute combinations (patch type / age class / patch size), then sort these in decreasing patch size before doing the spatial allocation. Allocating the smaller patches at the end of the sequence increases the likelihood that patch growth will reach target patch size before running out of adjacent unallocated cells.

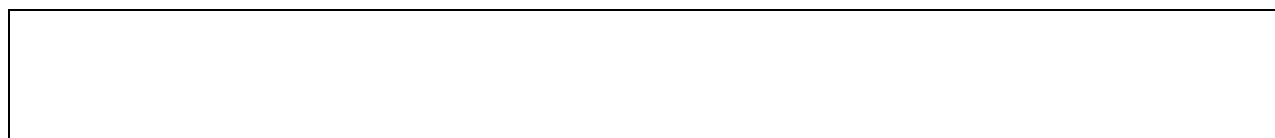
If the model produces adjacent patches with identical type and age attributes, the net result is one large patch produced from two adjacent identical patches. This will tend to increase the mean patch size. This can be avoided by querying neighbouring patches for identical attribute combinations prior to accepting patch growth attempt. Rejected samples can be returned to the sorted pool of unallocated samples.

Edge Effects

Patches that are generated along the edges of the model will be more likely to run out of room before growth has reached the target patch size. These edge effects can be minimized by generating a cell matrix that is larger than the user-defined model extent, and subsequently discarding the unwanted perimeter cells.

Patch Shape

Earlier versions of the FLG produced patches with very predictable geometric shapes if patch growth was unobstructed, due to the concentric layers of selected cells used to simulate patch growth. In an effort to increase patch shape variability, a subset of the selected layer of adjacent cells is



discarded at each iteration of the patch growth algorithm. The result is a much more natural patch shape distribution. An example of a forest landscape generated using the improved spatial allocation algorithm can be found in Figure 2.

Conclusion

A useful system for generating realistic forest landscapes was developed. Shortcomings identified in an earlier model were addressed, and the FLG has been seamlessly integrated into the existing ArcView Graphical User Interface. The system can distribute spatial attributes in landscapes to reflect attributes specified by the user. Integration with existing GIS technology allows for easy interfacing with forest resource allocation models. Input parameters are readily available to the user, or relatively easy to obtain. The result is a simple and robust software tool that allows the user to generate hypothetical GIS forest databases with patch size, patch type and age class distributions similar to those found in the baseline forest. It is hoped that this tool will assist researchers in testing, evaluating and comparing models that deal with spatial forest landscape management objectives.

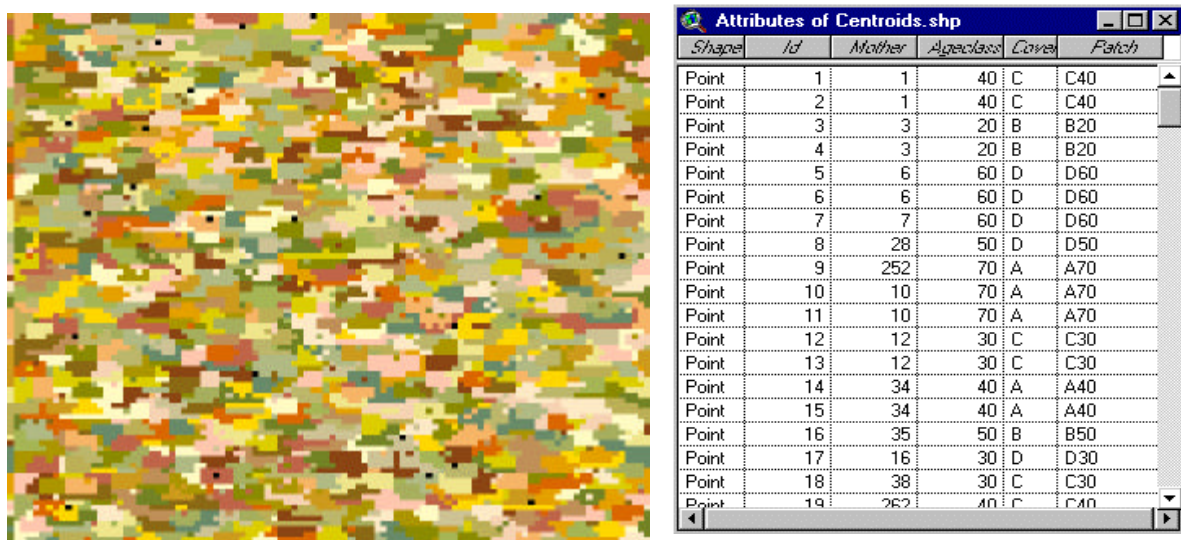


Figure 2: Sample raster cell matrix and cell attribute table produced by the FLG.